
Algorithm 1: SoundCode async verifier loop

Input: prompt x , model $\mathcal{M}$, verifier $\mathcal{V}$

Output: verified code y

```
1 code  $\leftarrow \emptyset$ ; ckpt_offsets  $\leftarrow [0]$ ; tasks  $\leftarrow []$ 
2 while not EOS do
3   tok  $\leftarrow$  LLM.next_token( $x$ , code)
4   code.append(tok)
   // Producer never blocks for the verifier
5   if at_structural_boundary(code) then
6     spawn async  $\tau \leftarrow \mathcal{V}.check(code)$            // non-blocking
7     tasks.append( $\tau$ )
   // Consumer drains finished verdicts each iteration
8   foreach  $\tau \in$  tasks and  $\tau.done$  do
9     if  $\tau.verdict = \textit{blocking}$  then
10      off  $\leftarrow$  ckpt_offsets  $[-2]$            // last verified checkpoint
11      code  $\leftarrow$  code  $[:off]$ 
12      LLM.set_position(off)
13      cancel_later_tasks()
14      break
15     else if  $\tau.verdict = \textit{ok}$  then
16      ckpt_offsets.append(|code|)           // advance frontier
17 return code
```
